

NUMERICAL ANALYSIS

GUIDED LAB 1

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1. Compute the absolute and relative error for

$$\sqrt{2} \approx 1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3}$$

Solution. Let's first compute the left and right hand side of the equation above using x_{approx} to represent the approximate solution (RHS). Using matlab we see that

$$\begin{aligned}x &\approx 1.41421356237, \text{ and} \\x_{\text{approx}} &= 1.4142129630\end{aligned}$$

Let ε_{abs} represent the absolute error, and ε_{rel} represent the relative error. Then we have the following:

$$\begin{aligned}\varepsilon_{\text{abs}} &= |\sqrt{2} - x_{\text{approx}}| \\&= |1.41421356237 - 1.4142129630| = 5.99 \times 10^{-7}\end{aligned}$$

and we also see that:

$$\begin{aligned}\varepsilon_{\text{rel}} &= \frac{\varepsilon_{\text{abs}}}{\sqrt{2}} \\&\approx \frac{5.99 \times 10^{-7}}{1.41421356237} \\&\approx 4.24 \times 10^{-7}\end{aligned}$$

□

2. The Newton-Raphason method for determining $\sqrt{a}$ follows from repeated iteration of

$$y_{n+1} = \frac{1}{2} \left(y_n + \frac{a}{y_n} \right)$$

Show that $y = \sqrt{a}$ is the fixed point of this sequence.

Proof. To achieve $y = \sqrt{a}$ as a fixed point of our equation we will need to compare the error between the n^{th} term and the $(n+1)^{\text{st}}$ term.

We begin by subtracting $\sqrt{a}$ from both sides of the equation and performing some algebraic manipulation. Consider

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) - \sqrt{a} \\ &= \frac{x_n^2 - 2x_n\sqrt{a} + a}{2x_n} \\ &= \frac{(x_n - \sqrt{a})^2}{2x_n} \end{aligned}$$

Accordingly,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{|x_{n+1} - \sqrt{a}|}{|x_n - \sqrt{a}|} &= \lim_{n \rightarrow \infty} \frac{1}{2x_n} \\ &= \frac{1}{2\sqrt{a}} \end{aligned}$$

Hence the sequence generated by this scheme has order of convergence equal to 2 and asymptotic error constant $1/(2\sqrt{a})$. $\square$

3. Create a script that implements Heron's algorithm (the formula in problem 1). It should take as the input: (1) the number a to square root, (2) the initial guess y_0 , and (3) the total number N of iterations. In addition to outputting the approximate square root at each iteration, you may use a built-in function as the "exact" answer for comparison. Plot both the relative error as a function of iteration for $N = 6$ on a log-log scale for $a = 2$. How sensitive is the result to your initial guess of y_0 ?

Solution. First note that the file <https://github.com/chtompki/MathSpring2026/tree/main/NumericalAna>

Consider a function that we have in file "problem3.m" that we have below:

```
function [p,absError,relError] = fixpt(a,y,n)
% Input - a, the number to square root
%       - y, the initial guess
%       - n, the number of iterations to run
p = y; % Initialize p with the initial guess
absError = zeros(1,n);
relError = zeros(1,n);
for k = 1:n
    p_new = 0.5 * (p + a / p);
    absError(k) = abs(p_new - sqrt(a));
    relError(k) = absError(k) / abs(p_new);
    if absError < 1e-10
        break;
    end
end
```

```

        end
        p = p_new;
    end
end

format long
[p,absError,relError] = fixpt(2,30,6)
loglog(1:6,absError, '-k')
hold on
loglog(1:6,relError,'--k')
hold off
legend("Absolute Error", "Relative Error")

```

So we see our plot from the above results as: But, if we fiddle with the initial value for the

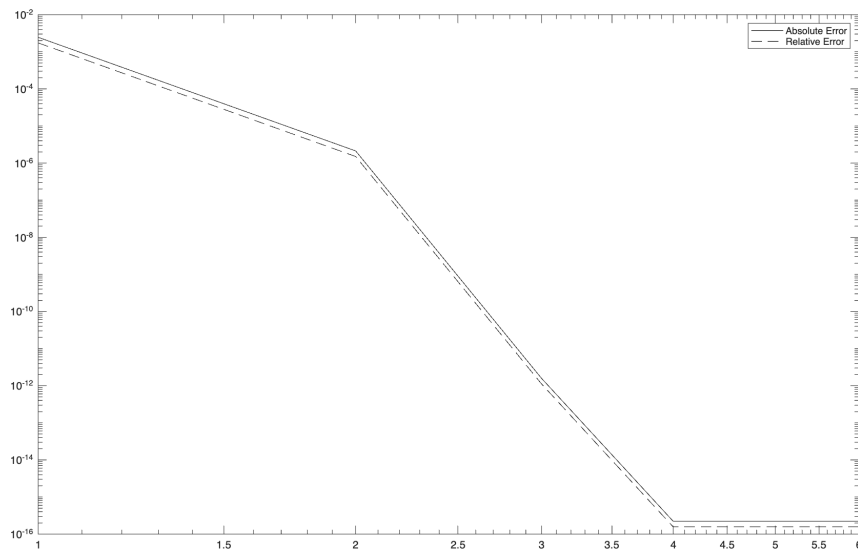


FIGURE 1. Log-Log plot of the absolute and relative error for Newton Raphson for $N = 6$, $y_0 = 1.5$

answer we see different errors. Consider Figure 2 which shows the errors when we set $y_0 = 30$ and we are done. □

4.

Solution. First note that the file <https://github.com/chtompki/MathSpring2026/tree/main/NumericalAna>

Compute the *approximate* relative error, and compare to the “true” errors. So we vary our algorithm a little as follows

```
function [p,absError,relError] = fixpt(a,y,n)
```

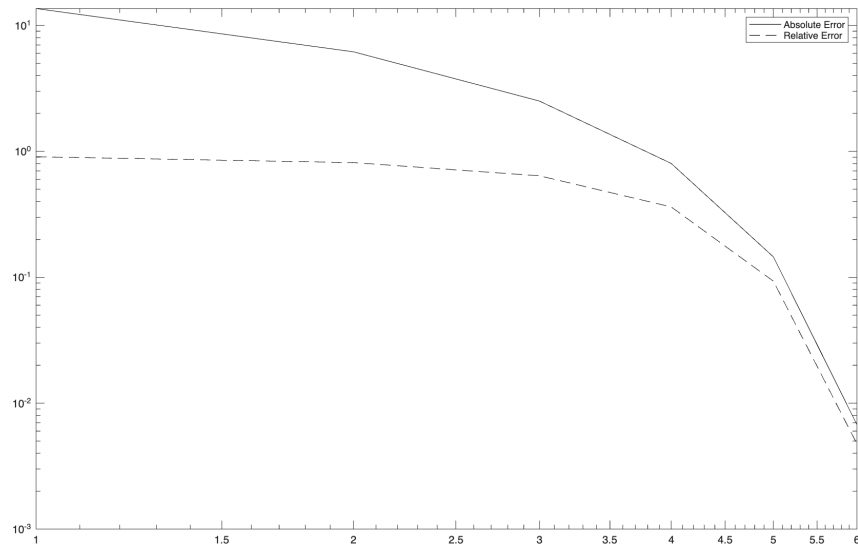


FIGURE 2. Log-Log plot of the absolute and relative error for Newton Raphson for $N = 6$, $y_0 = 30$

```
% Input - a, the number to square root
%       - y, the initial guess
%       - n, the number of iterations to run
p = y; % Initialize p with the initial guess
absError = zeros(1,n);
relError = zeros(1,n);
for k = 1:n
    p_new = 0.5 * (p + a / p);
    absError(k) = abs(p_new - p);
    relError(k) = absError(k) / abs(p_new);
    if absError < 1e-10
        break;
    end
    p = p_new;
end
end

format long
[p,absError,relError] = fixpt(2,30,6)
loglog(1:6,absError, '-k')
hold on
loglog(1:6,relError,'--k')
hold off
```

```
legend("Absolute Error", "Relative Error")
```

And for $N = 6$ and $y_0 = 30$ we see get the following errors

$$\varepsilon_6^{\text{abs}} = \begin{bmatrix} 14.966666666666667 \\ 7.450147819660015 \\ 3.659722054452168 \\ 1.706854881541622 \\ 0.657164646924791 \\ 0.138467746307099 \end{bmatrix}$$

and

$$\varepsilon_6^{\text{rel}} = \begin{bmatrix} 0.995565410199557 \\ 0.982456225846853 \\ 0.932778422047204 \\ 0.770029900059511 \\ 0.421409602464346 \\ 0.097445508110978 \end{bmatrix}$$

□

5. Plug in the equation $y_n = \sqrt{a}(1 + \delta_n)$ where $\varepsilon_n^{\text{rel}} = |\delta_n|$. Assuming that $|\delta_n| \ll 1$, simplify that to get that $\delta_{n+1} = C\delta_n^2 + \mathcal{O}(\delta_n^3)$.

Solution. Problem 5

□

6.

Solution. Problem 6

□

7.

Solution. Problem 7

□

8.

Solution. Problem 8

□

9.

Solution. Problem 9

□

10.

Solution. Problem 10

□